

Multiple Choice (10 marks)

1. A ball is released from rest at a certain height. What is its velocity after falling 256 ft?
 - (a) 64 ft/sec
 - (b) 160 ft/sec
 - (c) 80 ft/sec
 - (d) 256 ft/sec
 - (e) 128 ft/sec
2. When water changes to steam, energy is
 - (a) converted into matter
 - (b) changed to a different form
 - (c) transferred to the surrounding environment
 - (d) created within the water
 - (e) both absorbed and released simultaneously
3. Water stands 12.0 m deep in a storage tank whose top is open to the atmosphere. What are the gauge pressures at the bottom of the tank? (Unit: $10^5 Pa$)
 4. 1.60
 5. 1.22
 6. 0.98
 7. 1.45
 8. 1.05

A 4.0-gm bullet is fired from a 5.0-kg gun with a speed of 600 m/sec. What is the speed of recoil of the gun?

- (a) - 0.50 m/sec
- (b) - 0.52 m/sec
- (c) - 0.45 m/sec
- (d) - 0.38 m/sec
- (e) - 0.60 m/sec

The component of velocity that can remain constant for a tossed baseball is

- (a) Both horizontal and vertical
- (b) Diagonal
- (c) horizontal
- (d) Angular
- (e) Radial

True / False (10 marks)

Answer True or False

- 6. True or False: The lunar maria are dark lavas that fill older impact basins on the Moon's surface.
- 7. True or False: The temperature at which the population in the first excited state of I_2 is half that of the ground state is 390 K.
- 8. True or False: Hydrogen is the most common element found among the Jovian planets in our solar system.
- 9. True or False: Molecules in the atmosphere scatter blue light more effectively than red light, resulting in a blue sky.
- 10. True or False: In the diamond structure of elemental carbon, the nearest neighbors of each C atom are located at the corners of a triangle.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the concept of resultant force and calculate the resultant force acting on a body with a mass of 48 kg that is accelerating at 6 m/s^2 . Explain your reasoning and the underlying principles.
- 12. An astronaut weighing 800 N on Earth lands on a planet with double the mass and radius of Earth. Analyze the gravitational forces at play and calculate the astronaut's weight on this planet, providing a detailed explanation of your reasoning.

End of Paper